Principles of Quantum Mechanics

Part II

Recorded from the lecturer in 2026-2027.

0. Introduction

∗ Recall features from IB QM (Supplementary Notes 1)

- Wave-particle duality:

  Waves behaving like particles, e.g. light quanta, & vice versa (particles behaving like waves).

- Wavefunction $\psi(x)$ for a particle — prob. density $|\psi(x)|^2$.

  (Prob. intrinsic to theory.)

- Observables became operators on wavefns:

$$\hat{x} \to x, \qquad \hat{p} \to -i\hbar \frac{\partial}{\partial x}.$$

  & lack of commutativity, e.g.

$$[\hat{x}, \hat{p}] = i\hbar$$

  leads to Uncertainty Relation.

- Schrödinger eqn with Hamiltonian H specifies dynamics & eigenvalues of H are allowed energy levels of system.

- All of this explains H atom structure & resulting line spectra.

∗ Aims of this course:

- Reformulate QM in a more powerful, flexible, & useful form — the Dirac formalism (go beyond wavefns!).

Previous description appears as particular realisation & also leads to alternatives, e.g.

$$\tilde{\psi}(p) \qquad \text{with} \qquad \hat{x} \to +i\hbar \frac{\partial}{\partial p} \quad \& \quad \hat{p} \to p.$$

This more general approach allows

- Simpler analysis of known problems, e.g. oscillator

but also lots of new things

- Spin of particles
- Symmetries (translations, rotations, for example) & conservation laws.
- Identical particles (bosons & fermions)
- Framework for quantising more general systems, e.g. EM field
- Perturbation theory: practical tool for computations — we will not look at too many detailed applications, but we will try to keep track of what all this is for — physical predictions

1. Dirac Formalism

1.1 Concepts & Definitions

A quantum system is described at each instant by a state $|\psi\rangle$ which belongs to a complex vector space V; any non-zero element of this space V is a possible state.

Thus

$$|\psi\rangle \;\&\; |\phi\rangle \in V \quad \Rightarrow \quad \alpha|\psi\rangle + \beta|\phi\rangle \in V \qquad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour, but these states are not wavefns. There are also dual or conjugate states $\langle\phi|$ which belong to the dual space $V^\dagger$.

By definition, states & duals can be combined or paired to give a complex number

$$\underset{\text{"bra"}}{\langle\phi|}\,, \quad \underset{\text{"ket"}}{|\psi\rangle} \longmapsto \underset{\text{"bra(c)ket"}}{\langle\phi|\psi\rangle}\,, \qquad V^\dagger \times V \to \mathbb{C},$$

with

$$\langle\phi|\big(\alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle\big) = \alpha_1\langle\phi|\psi_1\rangle + \alpha_2\langle\phi|\psi_2\rangle,$$
$$\big(\beta_1\langle\phi_1| + \beta_2\langle\phi_2|\big)|\psi\rangle = \beta_1\langle\phi_1|\psi\rangle + \beta_2\langle\phi_2|\psi\rangle.$$

The spaces V & $V^\dagger$ come with an inner product, which can be described as a one-to-one correspondence between the spaces

$$V \longleftrightarrow V^\dagger,$$

with

$$|\psi\rangle \longleftrightarrow \langle\psi| = \big(|\psi\rangle\big)^\dagger \qquad \text{(use same symbol for corresponding states)}.$$

The inner product is

$$|\phi\rangle,\ |\psi\rangle \longmapsto \langle\phi|\psi\rangle = \big(|\phi\rangle\big)^\dagger|\psi\rangle, \qquad V \times V \to \mathbb{C},$$

& it is assumed to obey

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* \qquad \text{hermitian},$$
$$\big\|\,|\psi\rangle\,\big\|^2 = \langle\psi|\psi\rangle \qquad \text{real},$$
$$\left.\begin{array}{l}\langle\psi|\psi\rangle \geq 0,\\ \langle\psi|\psi\rangle = 0 \quad \text{iff} \quad |\psi\rangle = 0\end{array}\right\} \qquad \text{positive definite},$$

where now $\big\|\,|\psi\rangle\,\big\|$ is called the norm.

Note: Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ (& vice versa).

[Check:

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \quad \Rightarrow \quad \langle\phi|\chi\rangle = 0, \qquad \text{where } |\chi\rangle = |\psi_1\rangle - |\psi_2\rangle.$$

If this holds $\forall\,\langle\phi|$, set $\langle\phi| = \langle\chi|$ & get

$$\langle\chi|\chi\rangle = \big\|\,|\chi\rangle\,\big\|^2 = 0 \quad \Rightarrow \quad |\chi\rangle = 0, \qquad \text{as claimed.}$$

]

The physical content of any state is unchanged by

$$|\psi\rangle \to \alpha|\psi\rangle \qquad \text{for any } \alpha \in \mathbb{C}, \quad \alpha \neq 0.$$

We will usually normalise to make $\big\|\,|\psi\rangle\,\big\| = 1$, but can still change

$$|\psi\rangle \to e^{i\theta}|\psi\rangle.$$

Relative phases can be important in expressions like $\alpha|\psi\rangle + \beta|\phi\rangle$.

The space V is complete: we assume that appropriate sequences & series converge. A complete inner product space is called a Hilbert space.

An operator Q is a linear map on states

$$|\psi\rangle \longmapsto Q|\psi\rangle, \qquad V \to V,$$

with

$$Q\big(\alpha|\psi\rangle + \beta|\phi\rangle\big) = \alpha Q|\psi\rangle + \beta Q|\phi\rangle.$$

The same operator can be thought of as acting to the left on dual states,

$$\langle\phi| \longmapsto \langle\phi|Q, \qquad V^\dagger \to V^\dagger,$$

where the right-hand side is defined by

$$\big(\langle\phi|Q\big)|\psi\rangle = \langle\phi|\big(Q|\psi\rangle\big) = \langle\phi|Q|\psi\rangle.$$

End of lecture 1

Last time: abstract notation for states

$$|\psi\rangle \in V, \qquad \langle\phi| \in V^\dagger, \qquad \& \text{ map } V \longleftrightarrow V^\dagger,$$

& operators acting, e.g. $|\psi\rangle \longmapsto Q|\psi\rangle$ (§ 1.1).

Note: States & operators $|\psi\rangle$, $\langle\phi|$, Q can be dealt with by rules that mirror those for complex vectors $\underline{u}$ (column), $\underline{v}^\dagger$ (row), & a matrix A; e.g. compare

$$\langle\phi|Q|\psi\rangle \qquad \text{with} \qquad \underline{v}^\dagger A \underline{u}.$$

For any operator Q, say $|\psi\rangle \neq 0$ is an eigenstate with eigenvalue λ if

$$Q|\psi\rangle = \lambda|\psi\rangle;$$

equivalently,

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi|.$$

For general Q, λ may be complex.

For any Q, the hermitian conjugate or adjoint $Q^\dagger$ is defined by

$$\langle\phi|Q^\dagger = \big(Q|\phi\rangle\big)^\dagger \qquad \text{for any } |\phi\rangle.$$

Equivalently,

$$\begin{aligned}\langle\phi|Q^\dagger|\psi\rangle &= \big(Q|\phi\rangle\big)^\dagger|\psi\rangle \\ &= \langle\psi|Q|\phi\rangle^* \qquad \text{for all } |\phi\rangle,\ |\psi\rangle.\end{aligned}$$

Simple consequences:

$$\begin{aligned}(\alpha A + \beta B)^\dagger &= \alpha^* A^\dagger + \beta^* B^\dagger, \\ (AB)^\dagger &= B^\dagger A^\dagger,\end{aligned}$$

for any operators A, B & $\alpha, \beta \in \mathbb{C}$, &

$$\big(Q^\dagger\big)^\dagger = Q.$$

Observables

An operator Q is hermitian or self-adjoint if

$$Q^\dagger = Q;$$

such operators are called observables because they correspond to physical quantities that can be measured, e.g. position, momentum, energy.

To define a quantum system, we must specify the Hamiltonian H whose eigenvalues are the allowed energy levels of the system:

$$H|\chi\rangle = E|\chi\rangle \qquad \text{energy eigenvalue problem.}$$

Key results for any hermitian Q:

(i) All eigenvalues are real.

(ii) Eigenstates with distinct eigenvalues are orthogonal.

(iii) Any state can be expanded in terms of (written as a linear combination of) eigenstates.

We prove (i) & (ii) & assume (iii).

(i) Suppose

$$Q|\psi\rangle = \lambda|\psi\rangle.$$

Then

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi| = \langle\psi|Q \qquad (Q^\dagger = Q).$$

Hence

$$\langle\psi|Q|\psi\rangle = \lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle.$$

But

$$\big\| \, |\psi\rangle \, \big\|^2 = \langle\psi|\psi\rangle \neq 0 \qquad (|\psi\rangle \neq 0),$$

so $\lambda^* = \lambda$.

(ii) Let $|n\rangle$ be eigenstates of Q with eigenvalues $\lambda = q_n$ (real), where n is some discrete label, possibly ranging over infinitely many values.

$$Q|n\rangle = q_n|n\rangle, \qquad Q|m\rangle = q_m|m\rangle,$$

or

$$\langle m|Q = q_m\langle m| \qquad (q_m^* = q_m).$$

Hence

$$\langle m|Q|n\rangle = q_n\langle m|n\rangle = q_m\langle m|n\rangle.$$

But $q_m \neq q_n$, so $\langle m|n\rangle = 0$.

1.2 The Harmonic Oscillator

A one-dimensional harmonic oscillator of mass m & frequency ω is a system with position $\hat{x}$, momentum $\hat{p}$ & Hamiltonian

$$H = \frac{1}{2m}\hat{p}^{\,2} + \frac{1}{2}m\omega^2\hat{x}^{\,2}.$$

Seek eigenstates & eigenvalues of H.

(a) Annihilation, creation & number operators

Define

$$\begin{cases} a = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \\ a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right). \end{cases}$$

These are dimensionless hermitian conjugates, since $\hat{x}^\dagger = \hat{x}$, $\hat{p}^\dagger = \hat{p}$. Equivalently,

$$\begin{cases} \hat{x} = \left(\frac{\hbar}{2m\omega}\right)^{1/2}(a + a^\dagger), \\ \hat{p} = \left(\frac{\hbar m\omega}{2}\right)^{1/2} i(a^\dagger - a). \end{cases}$$

Easy to check:

$$[\hat{x}, \hat{p}] = i\hbar \quad \Longleftrightarrow \quad [a, a^\dagger] = 1.$$

In fact,

$$\begin{aligned} aa^\dagger &= \frac{m\omega}{2\hbar}\hat{x}^{\,2} + \frac{1}{2m\omega\hbar}\hat{p}^{\,2} - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}H + \frac{1}{2}, \end{aligned}$$

but

$$a^\dagger a = \frac{1}{\hbar\omega}H - \frac{1}{2}.$$

This confirms the commutation relation, but we also find

$$H = \hbar\omega\left(N + \frac{1}{2}\right),$$

with $N = a^\dagger a$ the number operator.

$N^\dagger = N$, & finding eigenvalues & eigenstates of N is equivalent to finding those for H. Let $|\lambda\rangle$ be any eigenstate of N with eigenvalue λ:

$$N|\lambda\rangle = \lambda|\lambda\rangle \quad \Longleftrightarrow \quad H|\lambda\rangle = E|\lambda\rangle, \qquad E = \hbar\omega\left(\lambda + \frac{1}{2}\right).$$

Then, assuming $\big\|\,|\lambda\rangle\,\big\|^2 = 1$ (normalised),

$$\begin{aligned} \lambda = \langle\lambda|N|\lambda\rangle &= \langle\lambda|a^\dagger a|\lambda\rangle = \big(a|\lambda\rangle\big)^\dagger\big(a|\lambda\rangle\big) \\ &= \big\|a|\lambda\rangle\big\|^2. \end{aligned}$$

Thus $\lambda \geq 0$ (all eigenvalues are non-negative), & $\lambda = 0$ iff $a|\lambda\rangle = 0$.

Now consider

$$\begin{aligned} N\big(a^\dagger|\lambda\rangle\big) &= \big([N, a^\dagger] + a^\dagger N\big)|\lambda\rangle \\ &= \big(a^\dagger + a^\dagger\lambda\big)|\lambda\rangle \\ &= (\lambda + 1)\big(a^\dagger|\lambda\rangle\big). \end{aligned}$$

Similarly,

$$\begin{aligned} N\big(a|\lambda\rangle\big) &= \big([N, a] + aN\big)|\lambda\rangle \\ &= (-a + a\lambda)|\lambda\rangle \\ &= (\lambda - 1)a|\lambda\rangle. \end{aligned}$$

Here we have used the results for commutators

$$[N, a^\dagger] = a^\dagger \qquad \big(= [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger]\big),$$

&

$$[N, a] = -a.$$

New states will be eigenstates provided their norms are non-zero. Compute:

$$\big\| a|\lambda\rangle \big\|^2 = \lambda,$$

$$\begin{aligned} \big\| a^\dagger|\lambda\rangle \big\|^2 &= \langle\lambda| a a^\dagger |\lambda\rangle = \langle\lambda| a^\dagger a + 1 |\lambda\rangle \\ &= \lambda + 1. \end{aligned}$$

Therefore $a^\dagger$ & a are called <u>creation</u> (or raising) & <u>annihilation</u> (or lowering) operators, respectively. Since all eigenvalues are non-negative, any non-integer λ would eventually lead to a negative eigenvalue under repeated lowering, which is impossible. Thus, λ must be a non-negative integer. Repeated lowering then reaches the ground state $|0\rangle$, where $a|0\rangle = 0$. Since the raising operator $a^\dagger$ can act indefinitely, all non-negative integers are eigenvalues. Hence,

$$\boxed{\lambda = 0, 1, 2, \ldots}$$

End of lecture 2

From calculations of norms above, we can choose normalised eigenstates $|n\rangle$ related by

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad a|n\rangle = \sqrt{n}\,|n-1\rangle.$$

Starting from any state $|n\rangle$ we can generate all others. In particular, starting from the ground state $|0\rangle$ characterised by

$$a|0\rangle = 0,$$

have normalised eigenstate

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle.$$

a, $a^\dagger$ also called <u>ladder operators</u>.

(b) <u>Importance of the oscillator</u>

- A smooth potential $V(x)$ with equilibrium point $x = x_0$ (so that $V'(x_0) = 0$) can be expanded

  $$V(x_0 + x) = V(x_0) + \frac{1}{2}V''(x_0)x^2 + O(x^3).$$

  So, if $x = x_0$ is <u>stable</u>, can set $V''(x_0) = m\omega^2$ & for displacements x not too large, expect system to be well-approximated by oscillator.

  e.g. diatomic molecules — quantisation of vibrational energies important in understanding internal energy & heat capacities — macroscopic consequences.

- More complicated systems can be analysed in terms of <u>normal modes</u>: coherent motion in which all degrees of freedom oscillate with same common frequency ω. General solution for <u>classical</u> oscillator is

  $$x(t) = Ae^{-i\omega t} + A^* e^{+i\omega t} \qquad A \text{ complex constant.}$$

Normal modes for a system with variables x_N are solutions of form

$$x_N(t) = Au_N e^{-i\omega t} + A^* u_N^* e^{+i\omega t}$$

with u_N complex vector specific to each normal frequency ω.

e.g. crystal with # atom $\mathcal{N} \sim 10^{23}$ — forces approximately elastic & N labels directions of motion for each atom, so $3 \times \mathcal{N}$ values.

or EM field — normal modes oscillations of electric & magnetic fields at each point in space

$$\underline{E}(\underline{x}, t) = A\underline{u}(\underline{x})e^{-i\omega t} + A^*\underline{u}(\underline{x})^* e^{+i\omega t}$$

for some complex vector field $\underline{u}(\underline{x})$. This is exact.

- QM: normal modes are independent oscillators — treat these as above. For each mode have

$$a(\omega) \quad \& \quad a(\omega)^\dagger.$$

Ground state $|0\rangle$ with

$$a(\omega)|0\rangle = 0.$$

Other states e.g. $a^\dagger(\omega)|0\rangle$ & find they have energy & momentum & behave like particles.

$$\left.\begin{matrix} a \text{ destroys} \\ a^\dagger \text{ creates} \end{matrix}\right\} \text{particles} \begin{cases} \text{phonons in crystal} \\ \text{photons in EM field} \end{cases}$$

1.3 Bases & Measurements

(a) Bases of Eigenstates

From key properties (i), (ii), (iii) in § 1.1 we know that for any observable Q there is an orthonormal basis of eigenstates $\{|n\rangle\}$ for V

$$Q|n\rangle = q_n|n\rangle, \qquad \langle m|n\rangle = \delta_{mn}.$$

For any state,

$$|\psi\rangle = \sum_n \alpha_n |n\rangle, \qquad \text{where} \quad \alpha_n = \langle n|\psi\rangle.$$

Note

$$\big\| \, |\psi\rangle \, \big\|^2 = 1 \quad \Longleftrightarrow \quad \begin{aligned} \langle\psi|\psi\rangle &= \left(\sum_m \alpha_m^* \langle m|\right)\left(\sum_n \alpha_n |n\rangle\right) \\ &= \sum_n |\alpha_n|^2 = 1. \end{aligned}$$

Note that we can write

$$1 = \sum_n |n\rangle\langle n| \qquad \text{“completeness relation”}$$

since on any state

$$\begin{aligned} |\psi\rangle &= \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle \\ &= \sum_n |n\rangle\big(\langle n|\psi\rangle\big), \end{aligned}$$

so holds for any orthonormal basis.

In above there may be several independent states with same eigenvalue λ. The eigenspace for a given eval is

$$V_\lambda = \big\{|\psi\rangle : Q|\psi\rangle = \lambda|\psi\rangle\big\}.$$

& note V_λ has basis $\big\{|n\rangle : q_n = \lambda\big\}$. The degeneracy of λ is the number of states in this basis, i.e. $\dim V_\lambda$, but note that λ is also called non-degenerate if degeneracy is 1.

End of lecture 3

(b) Functions of Operators

It is straightforward to define $f(Q)$ for some operator Q if f is a polynomial or power series (assuming convergence), e.g.

$$f(Q) = Q^3 + Q \qquad \text{or} \qquad f(Q) = \exp Q = \sum_{n=0}^{\infty} \frac{1}{n!} Q^n.$$

Important examples are

$$e^{-iHt/\hbar} \qquad \text{time evolution operator } (H \text{ Hamiltonian})$$

&

$$e^{-ia\hat{p}/\hbar} \qquad \text{translation operator (see later section of Chapter 1).}$$

More generally, for Q an observable with basis of eigenstates as in (a), we can define

$$f(Q)|n\rangle = f(q_n)|n\rangle$$

& then defined on any state, provided $f(q_n)$ makes sense, e.g.

$$\frac{1}{Q}|n\rangle = \frac{1}{q_n}|n\rangle \qquad \text{defined for } q_n \neq 0.$$

In addition, note that

$$Q = \sum_n q_n |n\rangle\langle n| \qquad \& \qquad f(Q) = \sum_n f(q_n)|n\rangle\langle n|.$$

Recall, introduced eigenspaces

$$V_\lambda = \big\{|\psi\rangle : Q|\psi\rangle = \lambda|\psi\rangle\big\}.$$

A projection operator P_λ onto this subspace, $V \to V_\lambda$, can be defined by its action on basis states:

$$P_\lambda|n\rangle = \begin{cases} |n\rangle & \text{if } q_n = \lambda, \\ 0 & \text{else.} \end{cases}$$

This can be written

$$P_\lambda = \sum_{n:\, q_n = \lambda} |n\rangle\langle n|.$$

(c) Measurement Axioms

Consider a measurement of Q (observable) when the system is in state $|\psi\rangle$. Then

- The result is an eigenvalue λ of Q.
- This result is obtained with probability

$$p(\lambda) = \big\|P_\lambda|\psi\rangle\big\|^2 = \sum_{n:\, q_n = \lambda} |\alpha_n|^2.$$

- Immediately after the measurement the state is

$$cP_\lambda|\psi\rangle = c\sum_{n:\, q_n=\lambda} \alpha_n|n\rangle,$$

  with c a constant for normalisation.

Often have case with λ non-degenerate ($\dim V_\lambda = 1$), eigenstate $|n\rangle$ unique (up to a phase), & then

- $p(\lambda) = |\alpha_n|^2 = |\langle n|\psi\rangle|^2$, & $\alpha_n = \langle n|\psi\rangle$ is called the amplitude.
- State after measurement is $|n\rangle$.

In general

$$\sum_\lambda p(\lambda) = \sum_n |\alpha_n|^2 = 1.$$

Example

System with space V with basis states (orthonormal)

$$|1\rangle, \quad |2\rangle, \quad |3\rangle,$$

& these are eigenstates of Q with evals

$$q_1 = q_2 = 0, \qquad q_3 = 1.$$

State of system

$$|\psi\rangle = \frac{1}{\sqrt{6}}\big(2|1\rangle + |2\rangle + |3\rangle\big).$$

Result	Probability of measuring	State afterwards
1	$\left\|\frac{1}{\sqrt{6}}\right\|^2 = \frac{1}{6}$	$\|3\rangle$
0	$\left\|\frac{2}{\sqrt{6}}\right\|^2 + \left\|\frac{1}{\sqrt{6}}\right\|^2 = \frac{5}{6}$	$\frac{1}{\sqrt{5}}\big(2\|1\rangle + \|2\rangle\big)$

Expectation value (mean) of Q in state $|\psi\rangle$ is

$$\langle Q\rangle_\psi = \langle\psi|Q|\psi\rangle = \sum_\lambda \lambda p(\lambda) = \sum_n q_n|\alpha_n|^2,$$

& the uncertainty (spread or variance) is

$$(\Delta Q)^2_\psi = \big\langle (Q - \langle Q\rangle_\psi)^2\big\rangle_\psi = \langle Q^2\rangle_\psi - \langle Q\rangle^2_\psi.$$

In thinking about many repeated measurements: must prepare the system in the same state $|\psi\rangle$ each time.

Note: If $|\psi\rangle$ is an eigenstate of Q with eval λ, then

$$\langle Q\rangle_\psi = \lambda, \qquad \& \qquad (\Delta Q)_\psi = 0.$$

QM behaviour arises from fact that certain observables do not commute. In general, in any state $|\psi\rangle$,

$$(\Delta A)_\psi(\Delta B)_\psi \geq \frac{1}{2}\big|\langle [A,B]\rangle_\psi\big|,$$

so $[A, B] \neq 0 \Rightarrow$ cannot expect to measure A & B precisely at same time.
[Result follows from

$$\left\| (A + i\lambda B)|\psi\rangle \right\|^2 \geq 0$$

for all real λ; expand & look at discriminant of quadratic in λ.]

Prototype case: position & momentum $\hat{x}$, $\hat{p}$ obey

$$[\hat{x}, \hat{p}] = i\hbar \qquad \text{in one dimension,} \qquad \& \qquad \Delta x \, \Delta p \geq \hbar/2.$$

In three dimensions have $\hat{x}_i$ & $\hat{p}_i$ with

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

& uncertainty applies to position/momentum in a given direction.

(d) Simultaneous measurement & complete commuting sets

If observable Q does not distinguish completely, because some evals are degenerate, then we can look for an additional observable Q' such that Q, Q' are simultaneously measurable. This means we have joint eigenstates $|\lambda, \lambda'\rangle$ with

$$Q|\lambda, \lambda'\rangle = \lambda|\lambda, \lambda'\rangle,$$
$$Q'|\lambda, \lambda'\rangle = \lambda'|\lambda, \lambda'\rangle.$$

This implies

$$QQ'|\lambda, \lambda'\rangle = Q'Q|\lambda, \lambda'\rangle.$$

Hence, if we have a basis of such eigenstates, then $QQ' = Q'Q$ on any state, or $[Q, Q'] = 0$. Conversely, if $[Q, Q'] = 0$, easy to check that

$$Q' : V_\lambda \to V_\lambda \qquad (V_\lambda \text{ eigenspace for } Q),$$

& finding a basis for V_λ consisting of eigenstates of Q' then gives us a basis of joint eigenstates.

End of lecture 4

§ 1.3 (d) — continued

A set of observables $\{Q, Q', Q'', \ldots\}$ is a complete commuting set if any pair commute $\Rightarrow$ a basis of joint eigenstates, & the set of eigenvalues, $\lambda, \lambda', \lambda'', \ldots$ label states uniquely (up to normalisation).

Equivalent, if A is any observable such that

$$[A, Q] = [A, Q'] = \cdots = 0$$

then

$$A = f(Q, Q', \ldots).$$

Example

Refer to example in (c): orthonormal states $|1\rangle$, $|2\rangle$, $|3\rangle$ eigenstates of Q with $\lambda(= q_1 = q_2) = 0$, $\lambda(= q_3) = 1$.

Second observable Q' with

$$Q'|1\rangle = |2\rangle,$$
$$Q'|2\rangle = |1\rangle,$$
$$Q'|3\rangle = 0.$$

Check

$$[Q, Q'] = 0 \qquad (Q' : V_{\lambda=0} \to V_{\lambda=0}).$$

Let

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|1\rangle \pm |2\rangle\big)$$

then

$$Q'|\pm\rangle = \pm|\pm\rangle.$$

Q, Q' complete commuting set with basis of joint eigenstates

	λ	λ'
$\|+\rangle$	0	1
$\|-\rangle$	0	-1
$\|3\rangle$	1	0

1.4 Bases, Representations & Wavefunctions

(a) Discrete & Continuous Bases

Consider orthonormal basis $\{|n\rangle\}$ with n discrete label. We can regard all states & operators as row/col vecs & matrices, respectively:

$$|\psi\rangle = \sum_n \alpha_n |n\rangle \quad \Longleftrightarrow \quad \alpha_n = \langle n|\psi\rangle,$$
$$|\phi\rangle = \sum_n \beta_n |n\rangle \quad \Longleftrightarrow \quad \beta_n = \langle n|\phi\rangle.$$

cmpts of complex vectors

Inner product

$$\langle\phi|\psi\rangle = \sum_n \beta_n^* \alpha_n.$$

For an operator A,

$$A|n\rangle = \sum_m |m\rangle A_{mn}, \qquad \text{complex matrix } A_{mn} = \langle m|A|n\rangle.$$

Simple to check that

$$|\phi\rangle = A|\psi\rangle \quad \Longleftrightarrow \quad \beta_m = \sum_n A_{mn}\alpha_n$$

& also, if B is another operator, matrix $B_{mn} = \langle m|B|n\rangle$, then

$$(AB)_{mn} = \sum_p A_{mp}B_{pn} \qquad \text{(matrix multiplication)}.$$

The choice of basis provides a representation of the quantum states & operators.

We can also allow the index n to be continuous. This requires modifications

$$\sum_n \longrightarrow \int dn$$

$$\langle n|m\rangle = \delta_{nm} \longrightarrow \delta(n-m)$$

$$|\psi\rangle = \sum_n \alpha_n|n\rangle \longrightarrow \int dn\, \alpha_n|n\rangle$$

$$\alpha_n = \langle n|\psi\rangle.$$

Completeness relation

$$1 = \sum_n |n\rangle\langle n| \longrightarrow \int dn\, |n\rangle\langle n|.$$

Note: if basis states above arise as eigenstates of observable, then

$$|\alpha_n|^2 = |\langle n|\psi\rangle|^2 \longrightarrow \text{ probability density for continuous range of } n.$$

(b) Position & Momentum Bases — Wavefunctions

Consider a particle in one dimension, position $\hat{x}$ & momentum $\hat{p}$ obeying

$$[\hat{x}, \hat{p}] = i\hbar.$$

Let $|x\rangle$ be position eigenstates

$$\hat{x}|x\rangle = x|x\rangle$$

continuous evals with

$$\langle x|x'\rangle = \delta(x - x'), \qquad \int dx\, |x\rangle\langle x| = 1.$$

Any state can be written

$$|\psi\rangle = \int dx\, \psi(x)|x\rangle$$

with

$$\psi(x) = \langle x|\psi\rangle$$

& this is the usual position space wavefunction. From remarks in (a), $|\psi(x)|^2$ is a probability density for measuring position.

Inner product

$$\begin{aligned}\langle\phi|\psi\rangle &= \left(\int dx\, \phi(x)^*\langle x|\right)\left(\int dx'\, \psi(x')|x'\rangle\right) \\ &= \int dx \int dx'\, \phi(x)^*\psi(x') \underbrace{\langle x|x'\rangle}_{\delta(x-x')} = \int dx\, \phi(x)^*\psi(x)\end{aligned}$$

&

$$\big\|\, |\psi\rangle \,\big\|^2 = \langle\psi|\psi\rangle = \int dx\, |\psi(x)|^2 = 1, \quad \text{if normalised.}$$

Similarly, define momentum eigenstates $|p\rangle$ with

$$\hat{p}|p\rangle = p|p\rangle$$

&

$$\langle p|p'\rangle = \delta(p - p'), \qquad \int dp\, |p\rangle\langle p| = 1.$$

Now can expand

$$|\psi\rangle = \int dp\, \tilde{\psi}(p)|p\rangle$$

with

$$\tilde{\psi}(p) = \langle p|\psi\rangle$$

momentum space wavefunction & $|\tilde{\psi}(p)|^2$ probability distribution for measurements of momentum.

To proceed, we need

$$\left.\begin{aligned}\langle x|p\rangle &= \frac{1}{(2\pi\hbar)^{1/2}} e^{ipx/\hbar}, \\ \langle p|x\rangle &= \frac{1}{(2\pi\hbar)^{1/2}} e^{-ipx/\hbar}.\end{aligned}\right\} \qquad (*)$$

Assume $(*)$ holds & verify it below.

Consider action of $\hat{x}$ & $\hat{p}$ on position space wavefunctions

$$\hat{x}|\psi\rangle \quad \text{wavefn}: \quad \langle x|\hat{x}|\psi\rangle = x\langle x|\psi\rangle = x\psi(x).$$

$$\begin{aligned}\hat{p}|\psi\rangle \quad \text{wavefn}: \quad \langle x|\hat{p}|\psi\rangle & \\ &= \int dp\, \langle x|\hat{p}|p\rangle\langle p|\psi\rangle \qquad \text{completeness} \\ &= \int dp\, p\langle x|p\rangle\langle p|\psi\rangle.\end{aligned}$$

Note

$$p\langle x|p\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle \qquad \text{from } (*).$$

Hence

$$\begin{aligned}\langle x|\hat{p}|\psi\rangle &= -i\hbar\frac{\partial}{\partial x}\int dp\, \langle x|p\rangle\langle p|\psi\rangle \\ &= -i\hbar\frac{\partial}{\partial x}\langle x|\psi\rangle \qquad \text{completeness again} \\ &= -i\hbar\psi'(x).\end{aligned}$$

i.e.

$$\left.\begin{aligned}\hat{x} &\to x \\ \hat{p} &\to -i\hbar\,\partial/\partial x\end{aligned}\right\} \quad \text{on position space wavefunctions.}$$

Similarly, on momentum space wavefunctions, calculate

$$\begin{matrix}\langle p|\hat{x}|\psi\rangle \\ \langle p|\hat{p}|\psi\rangle\end{matrix} \qquad \text{\& find} \qquad \begin{matrix}\hat{x} \to i\hbar\dfrac{\partial}{\partial p} \\ \hat{p} \to p.\end{matrix}$$

Relationship between wavefunctions $\psi(x)$ & $\tilde{\psi}(p)$ also follows from $(*)$

$$\begin{aligned}\tilde{\psi}(p) = \langle p|\psi\rangle &= \int dx\, \langle p|x\rangle\langle x|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}}\int dx\, e^{-ipx/\hbar}\psi(x)\end{aligned}$$

&

$$\begin{aligned}\psi(x) = \langle x|\psi\rangle &= \int dp\, \langle x|p\rangle\langle p|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}}\int dp\, e^{ixp/\hbar}\tilde{\psi}(p).\end{aligned}$$

Fourier transform & inverse.

End of lecture 5

Now given $H(\hat{x},\hat{p})$ we can consider

$$H|\psi\rangle = E|\psi\rangle.$$

If

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\text{on } \psi(x) \qquad H \to -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x),$$
$$\text{on } \tilde{\psi}(p) \qquad H \to \frac{p^2}{2m} + V\left(i\hbar\frac{\partial}{\partial p}\right).$$

It may be easy to interpret potential term in momentum space. e.g. $V(x) = \lambda x^n$ then

$$V\left(i\hbar\frac{\partial}{\partial p}\right) = \lambda(i\hbar)^n\left(\frac{\partial}{\partial p}\right)^n$$

for n a positive integer. More generally, argue from first principles as follows.

Consider

$$\begin{aligned}
\langle p|V(\hat{x})|\psi\rangle &= \int dx\, \langle p|V(\hat{x})|x\rangle\langle x|\psi\rangle \\
&= \int dx \int dp'\, \langle p|V(\hat{x})|x\rangle\langle x|p'\rangle\langle p'|\psi\rangle \\
&= \int dp' \left(\int dx\, V(x) \underbrace{\langle p|x\rangle\langle x|p'\rangle}_{\text{use } (*)}\langle p'|\psi\rangle \right) \\
&= \int dp' \int dx\, V(x) \frac{e^{-i(p-p')x/\hbar}}{2\pi\hbar}\tilde{\psi}(p') \\
&= \frac{1}{(2\pi\hbar)^{1/2}} \int dp'\, \tilde{V}(p-p')\tilde{\psi}(p'),
\end{aligned}$$

reproducing convolution theorem.

i.e. $H|\psi\rangle = E|\psi\rangle$ becomes

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi \qquad \text{in position space}$$

or

$$\frac{p^2}{2m}\tilde{\psi}(p) + \frac{1}{(2\pi\hbar)^{1/2}} \int dp'\, \tilde{V}(p-p')\tilde{\psi}(p') = E\tilde{\psi}(p)$$

in momentum space.

To generalise from one to three dimensions: position & momentum operators obey

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}.$$

One complete commuting set is

$$\underline{\hat{x}} = (\hat{x}_1, \hat{x}_2, \hat{x}_3)$$

with joint eigenstates

$$\hat{x}_i|\underline{x}\rangle = x_i|\underline{x}\rangle$$

orthonormal

$$\langle \underline{x}|\underline{x}'\rangle = \delta^{(3)}(\underline{x} - \underline{x}').$$

An alternative set

$$\underline{\hat{p}} = (\hat{p}_1, \hat{p}_2, \hat{p}_3)$$

with joint eigenstates

$$\hat{p}_i|\underline{p}\rangle = p_i|\underline{p}\rangle \qquad \text{(also orthonormal)}.$$

For any state $|\psi\rangle$ we can expand

$$|\psi\rangle = \int d^3\underline{x}\, \psi(\underline{x})|\underline{x}\rangle = \int d^3\underline{p}\, \tilde{\psi}(\underline{p})|\underline{p}\rangle$$

&

$$\psi(\underline{x}) = \langle \underline{x}|\psi\rangle, \qquad \tilde{\psi}(\underline{p}) = \langle \underline{p}|\psi\rangle.$$

$(*)$ becomes

$$\langle \underline{x}|\underline{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\underline{p}\cdot\underline{x}/\hbar}$$

& wavefunctions related by

$$\tilde{\psi}(\underline{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\underline{x}\, e^{-i\underline{p}\cdot\underline{x}/\hbar}\psi(\underline{x}) \qquad (\& \text{ inverse}).$$

(c) More on position & momentum eigenstates

Return to one dimension & seek to justify $(*)$ using only $[\hat{x},\hat{p}] = i\hbar$.

Given $|x\rangle$ with $\hat{x}|x\rangle = x|x\rangle$, consider

$$T(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{-ia}{\hbar}\right)^n \hat{p}^n \qquad a \text{ any real number}$$

& note

$$[\hat{x}, T(a)] = \left[\hat{x}, \frac{-ia\hat{p}}{\hbar}\right] T(a) = aT(a) \qquad [\text{see 3 on Example Sheet 1}].$$

Hence

$$\begin{aligned}\hat{x}\big(T(a)|x\rangle\big) &= \big([\hat{x}, T(a)] + T(a)\hat{x}\big)|x\rangle \\ &= (x+a)\big(T(a)|x\rangle\big).\end{aligned}$$

Think of $T(a)|x\rangle$ as a translated eigenstate.

Similarly, consider

$$S(b) = e^{ib\hat{x}/\hbar}$$

& note

$$[\hat{p}, S(b)] = bS(b).$$

Hence

$$\hat{p}|p\rangle = p|p\rangle \quad \Rightarrow \quad \hat{p}\big(S(b)|p\rangle\big) = (p+b)\big(S(b)|p\rangle\big).$$

$S(b)$ gives translation in momentum space.

Now take fixed values x_0 & p_0 & define all eigenstates by

$$|x_0 + a\rangle = T(a)|x_0\rangle, \qquad |p_0 + b\rangle = S(b)|p_0\rangle.$$

Now observe

$$\begin{aligned}\langle x_0 + a|p_0 + b\rangle &= \langle x_0|e^{ia\hat{p}/\hbar}|p_0 + b\rangle \\ &= \langle x_0|e^{ia(p_0+b)/\hbar}e^{ib\hat{x}/\hbar}|p_0\rangle \\ &= e^{i(ab+ap_0+bx_0)/\hbar}\underbrace{\langle x_0|p_0\rangle}_{\text{choose } = \frac{1}{\sqrt{2\pi\hbar}}}.\end{aligned}$$

for correct, consistent normalisation.

Then set $x_0 = p_0 = 0$ to get

$$\langle a|b\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{iab/\hbar} \qquad \text{which is } (*).$$

2. Evolution in Time & Pictures

2.1 The Schrödinger Equation & the Schrödinger Picture

The evolution in time of a state $|\psi(t)\rangle$ is governed (in absence of external influences or measurements) by the Schrödinger equation (SE)

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle$$

where $H = H^\dagger$ is the Hamiltonian.

Equivalently

$$-i\hbar\frac{\partial}{\partial t}\langle\psi(t)| = \langle\psi(t)|H.$$

This implies

$$\frac{\partial}{\partial t}\big(\langle\psi(t)|\psi(t)\rangle\big) = 0$$

so that normalisation of $|\psi(t)\rangle$ is preserved in time.

Solution of SE can be written

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where

$U(t) = e^{-iHt/\hbar}$ time evolution operator, assuming H independent of t.

Check immediately from

$$i\hbar\frac{\partial U(t)}{\partial t} = HU(t).$$

Recall that time-evolution problems can also be conveniently solved by working with basis of energy eigenstates $\{|n\rangle\}$ with

$$H|n\rangle = E_n|n\rangle$$

with

$e^{-iE_nt/\hbar}|n\rangle$ solution of (t-dep) SE.

End of lecture 6

§ 2.1 Sch. eqn & Sch picture

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi\rangle$$
$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where

$$U(t) = e^{-iHt/\hbar} \qquad (H \text{ indep of } t)$$
$$= \sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{it}{\hbar}H\right)^n$$

If

$$|\psi(0)\rangle = \sum_n \alpha_n|n\rangle$$

where

$$H|n\rangle = E_n|n\rangle$$

then

$$|\psi(t)\rangle = \sum_n \alpha_n e^{-iE_nt/\hbar}|n\rangle$$

Example system with two energy eigenstates

$$\begin{array}{cc} |1\rangle & |2\rangle \\ E_1 & E_2 \end{array}$$

Consider measuring Q defined by

$$Q|1\rangle = |2\rangle, \qquad Q|2\rangle = |1\rangle$$

So

$$[Q, H] \neq 0$$

Eigenstates of Q:

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|1\rangle \pm |2\rangle\big) \qquad \text{evals } q_\pm = \pm 1$$

Take initial state $|\psi(0)\rangle = |+\rangle$

$$\Rightarrow \qquad |\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-iE_1 t/\hbar}|1\rangle + e^{-iE_2 t/\hbar}|2\rangle\right)$$

Probability of measuring Q at time t & getting ± 1 is

$$\begin{aligned} &|\langle \pm|\psi(t)\rangle|^2 \\ &= \left|\frac{1}{2}(\langle 1| \pm \langle 2|)\left(e^{-iE_1 t/\hbar}|1\rangle + e^{-iE_2 t/\hbar}|2\rangle\right)\right|^2 \\ &= \left|\frac{1}{2}\left(e^{-iE_1 t/\hbar} \pm e^{-iE_2 t/\hbar}\right)\right|^2 = \begin{cases} \cos^2\big((E_1 - E_2)t/2\hbar\big) \\ \sin^2\big((E_1 - E_2)t/2\hbar\big) \end{cases} \end{aligned}$$

In basis

$$\begin{aligned} |1\rangle &\to \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |2\rangle &\to \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned} \qquad H \to \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}$$

$$U = e^{-iHt/\hbar} \quad \to \quad \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 \\ 0 & e^{-iE_2 t/\hbar} \end{pmatrix}$$

In alternative basis

$$|+\rangle \to \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|-\rangle \to \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$H|\pm\rangle = \frac{1}{\sqrt{2}}\big(E_1|1\rangle + E_2|2\rangle\big)$$

$$\left.\begin{aligned} \langle \pm|H|\pm\rangle &= \frac{1}{2}(E_1 + E_2) = \mathcal{E} \\ \langle \mp|H|\pm\rangle &= \frac{1}{2}(E_1 - E_2) = \hbar\lambda \end{aligned}\right\} \quad \text{def } \mathcal{E} \text{ \& } \lambda \text{ consts.}$$

$$H \to \begin{pmatrix} \mathcal{E} & \hbar\lambda \\ \hbar\lambda & \mathcal{E} \end{pmatrix} = \mathcal{E} I + \hbar\lambda M$$

$$I \quad 2 \times 2 \text{ identity} \qquad \& \qquad M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Now

$$U(t) = \exp(-iHt/\hbar) \to e^{-\frac{i}{\hbar}t(\mathcal{E}I+\lambda\hbar M)}$$
$$= e^{-\frac{i}{\hbar}t\mathcal{E}}e^{-it\lambda M}$$

Note $M^2 = I$ & hence

$$\exp(-it\lambda M) = \sum_{r=0}^{\infty}\frac{1}{(2r)!}(-i\lambda t)^{2r}I + \sum_{r=0}^{\infty}\frac{1}{(2r+1)!}(-i\lambda t)^{2r+1}M$$
$$= (\cos\lambda t)I - i(\sin\lambda t)M.$$

$$|\psi(0)\rangle = |+\rangle \to \begin{pmatrix}1\\0\end{pmatrix}$$

$$|\psi(t)\rangle = e^{-iHt/\hbar}|+\rangle \to \left(e^{-it\mathcal{E}/\hbar}\right)\begin{pmatrix}\cos\lambda t\\-i\sin\lambda t\end{pmatrix}$$

$$e^{-it\mathcal{E}/\hbar}\big(\cos\lambda t\,|+\rangle - i\sin\lambda t\,|-\rangle\big) \qquad \longleftarrow$$

Now compare with previous results for prob. of measuring Q & finding ± 1

2.2 Heisenberg Picture & Heisenberg Equations

(a) Unitary operators

Physical predictions in QM are given by probability amplitudes, eigenvalues, expectation values, etc., hence by expressions like

$$\langle\phi|\psi\rangle, \quad \langle\phi|A|\psi\rangle, \quad \text{etc.}$$

An operator U is called unitary if

$$UU^\dagger = U^\dagger U = 1 \quad \text{or} \quad U^\dagger = U^{-1}.$$

Given any such U, consider a map on states & operators

$$|\psi\rangle \longrightarrow |\psi'\rangle = U|\psi\rangle$$
$$\langle\psi| \longrightarrow \langle\psi'| = \langle\psi|U^\dagger$$
$$A \longrightarrow A' = UAU^\dagger$$

Under this, all physical predictions remain unchanged

$$\text{e.g.} \qquad \langle\phi|\psi\rangle \to \langle\phi'|\psi'\rangle = \langle\phi|U^\dagger U|\psi\rangle = \langle\phi|\psi\rangle$$
$$\langle\phi|A|\psi\rangle \to \langle\phi'|A'|\psi'\rangle = \langle\phi|U^\dagger UAU^\dagger U|\psi\rangle$$
$$= \langle\phi|A|\psi\rangle.$$

Furthermore

$$C = AB \longrightarrow C' = A'B'$$
$$C = [A,B] \longrightarrow C' = [A',B']$$
$$Q^\dagger = Q \quad \Rightarrow \quad Q' = UQU^\dagger \quad \text{also satisfies } (Q')^\dagger = Q'$$
$$Q|\psi\rangle = \lambda|\psi\rangle \quad \Rightarrow \quad Q'|\psi'\rangle = \lambda|\psi'\rangle$$
$$\text{same eigenvalue.}$$

(b) Changing Pictures

In the Schrödinger picture

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle \qquad \begin{pmatrix}\text{all for } H \text{ indep of } t \\ \& \ H = H^\dagger\end{pmatrix}$$

$$\Rightarrow \qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where $U(t) = \exp(-iHt/\hbar)$ is unitary. In fact

$$U(t)^\dagger = U(t)^{-1} = U(-t)$$

$$\& \qquad U(t_1)U(t_2) = U(t_1 + t_2)$$

The states depend on t but observables do not, in S-picture.

To pass to Heisenberg picture we shift t-dependence from states to operators as follows

	Schrödinger	Heisenberg
states	$\|\psi(t)\rangle_S$	$\|\psi\rangle_H = U^\dagger\|\psi(t)\rangle_S = e^{itH/\hbar}\|\psi(t)\rangle_S = \|\psi(0)\rangle_S$
ops	A_S	$A_H(t) = U^\dagger A_S U = e^{itH/\hbar}A_S e^{-itH/\hbar}$

Because the transformation $U(t)$ is unitary

$${}_S\langle\phi(t)|A_S|\psi(t)\rangle_S = {}_H\langle\phi|A_H(t)|\psi\rangle_H$$

Note

$$H_H = H_S = H$$

To specify dynamics in H-picture, consider

$$\begin{aligned}\frac{d}{dt}A_H(t) &= \frac{d}{dt}\left(e^{iHt/\hbar}A_S e^{-iHt/\hbar}\right) \\ &= \frac{iH}{\hbar}e^{iHt/\hbar}A_S e^{-iHt/\hbar} - e^{iHt/\hbar}A_S e^{-iHt/\hbar}\frac{iH}{\hbar} \\ &= \frac{i}{\hbar}[H, A_H(t)]\end{aligned}$$

or

$$i\hbar\frac{d}{dt}A_H(t) = [A_H(t), H] \qquad \text{Heisenberg equations of motion}$$

e.g. particle in one dimension $\hat{x}(t)$, $\hat{p}(t)$ in H-picture

consider

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{i\hbar}[\hat{x}(t), H] = \frac{1}{m}\hat{p}(t)$$

$$\frac{d}{dt}\hat{p}(t) = \frac{1}{i\hbar}[\hat{p}(t), H] = -V'(\hat{x}(t))$$

using

$$[\hat{x}(t), \hat{p}(t)] = i\hbar$$

End of lecture 7

Taking expectation values in any state $|\psi\rangle$ (independent of t in the Heisenberg picture) gives

$$\frac{d}{dt}\langle\hat{x}\rangle = \frac{1}{m}\langle\hat{p}\rangle, \qquad \frac{d}{dt}\langle\hat{p}\rangle = -\langle V'(\hat{x})\rangle.$$

These are Ehrenfest's theorem, true in all pictures.

For some potentials we can solve the Heisenberg equations explicitly.

Free particle: $V = 0$

$$\frac{d}{dt}\hat{p}(t) = 0 \quad \Rightarrow \quad \hat{p}(t) = \hat{p}(0) \qquad \text{(constant operator)}.$$

Then

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{m}\hat{p}(0) \quad \Rightarrow \quad \hat{x}(t) = \hat{x}(0) + \frac{\hat{p}(0)}{m}t,$$

just as in classical dynamics.

Harmonic oscillator: $V = \frac{1}{2}m\omega^2 x^2$

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{m}\hat{p}(t), \qquad \frac{d}{dt}\hat{p}(t) = -m\omega^2\hat{x}(t).$$

Hence

$$\frac{d^2}{dt^2}\hat{x}(t) + \omega^2\hat{x}(t) = 0, \qquad \frac{d^2}{dt^2}\hat{p}(t) + \omega^2\hat{p}(t) = 0.$$

The solutions are

$$\hat{x}(t) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t,$$
$$\hat{p}(t) = \hat{p}(0)\cos\omega t - m\omega\hat{x}(0)\sin\omega t.$$

Or equivalently,

$$\hat{x}(t) = \left(\frac{\hbar}{2m\omega}\right)^{1/2}\left(ae^{-i\omega t} + a^\dagger e^{i\omega t}\right),$$
$$\hat{p}(t) = \left(\frac{\hbar m\omega}{2}\right)^{1/2}\frac{1}{i}\left(ae^{-i\omega t} - a^\dagger e^{i\omega t}\right),$$

where a, $a^\dagger$ are defined from the Schrödinger-picture operators $\hat{x}(0)$, $\hat{p}(0)$.

Why do the ladder operators acquire these phases?

For the harmonic oscillator,

$$H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right), \qquad [a, a^\dagger] = 1.$$

Using $[AB, C] = A[B, C] + [A, C]B$, we find

$$[a^\dagger a, a] = a^\dagger[a, a] + [a^\dagger, a]a = -a,$$
$$[a^\dagger a, a^\dagger] = a^\dagger[a, a^\dagger] + [a^\dagger, a^\dagger]a = a^\dagger.$$

The constant term in H commutes with every operator, so

$$[H, a] = -\hbar\omega a, \qquad [H, a^\dagger] = \hbar\omega a^\dagger.$$

Define

$$a_H(t) = U^\dagger(t)aU(t), \qquad a_H^\dagger(t) = U^\dagger(t)a^\dagger U(t).$$

Since H commutes with $U(t)$, the same commutator relations hold for the Heisenberg-picture operators. Their equations of motion are therefore

$$\frac{d}{dt}a_H(t) = \frac{i}{\hbar}[H, a_H(t)] = -i\omega a_H(t),$$
$$\frac{d}{dt}a_H^\dagger(t) = \frac{i}{\hbar}[H, a_H^\dagger(t)] = i\omega a_H^\dagger(t).$$

With $a_H(0) = a$ and $a_H^\dagger(0) = a^\dagger$, this gives

$$\boxed{a_H(t) = ae^{-i\omega t}, \qquad a_H^\dagger(t) = a^\dagger e^{i\omega t}.}$$

The two phase factors have opposite signs because the commutators of H with the lowering and raising operators have opposite signs. The second result also follows by taking the adjoint of the first. Substituting these results into

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}}\big(a_H(t) + a_H^\dagger(t)\big), \qquad \hat{p}(t) = \frac{1}{i}\sqrt{\frac{\hbar m\omega}{2}}\big(a_H(t) - a_H^\dagger(t)\big)$$

recovers the expressions above. Here a and $a^\dagger$ without a time argument denote the fixed operators at $t = 0$.

2.3 Canonical Quantisation

$*$ Non-examinable material

This is the final step in Dirac's approach to constructing QM: how does one pass from a general classical system to its quantum version?

A classical system can be described by a set of generalised positions $x_i(t)$ and momenta $p_i(t)$, $1 \le i \le N$ (which may include angles and angular momenta). These are coordinates on phase space. The dynamics are governed by a Hamiltonian $H(x_i, p_i)$.

The Poisson bracket of two functions $f(x_i, p_i)$, $g(x_i, p_i)$ is defined by

$$\{f, g\} = \sum_{i=1}^{N}\left(\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial x_i}\right),$$

which is another function on phase space. In particular,

$$\{x_i, p_j\} = \delta_{ij}.$$

Properties of Poisson brackets closely resemble those of commutators (linearity, the Jacobi identity, etc.). Dynamics in this formulation are given by Hamilton's equation

$$\frac{df}{dt} = \{f, H\} \qquad \text{for } f \text{ with no explicit time dependence.}$$

In canonical quantisation, classical functions are promoted to quantum operators,

$$f, g \quad \longrightarrow \quad \hat{f}, \hat{g},$$

and Poisson brackets determine commutators through Dirac's formal rule

$$[\hat{f}, \hat{g}] = i\hbar\widehat{\{f, g\}}.$$

In particular, this recovers

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

and the correspondence with Hamilton's equation gives the Heisenberg equation

$$i\hbar\frac{d\hat{f}}{dt} = [\hat{f}, H].$$

The rule is a quantisation prescription; for general functions, operator ordering must be specified and the rule cannot hold exactly for every pair of functions simultaneously.

$*$ End of non-examinable material

Tensor Products

Given state spaces V_1 and V_2, their tensor product $V = V_1 \otimes V_2$ is spanned by tensor product states

$$|\psi\rangle \otimes |\phi\rangle, \qquad |\psi\rangle \in V_1, \quad |\phi\rangle \in V_2,$$

with duals $\langle\psi| \otimes \langle\phi|$, subject to

$$\begin{aligned}
(|\psi\rangle + |\psi'\rangle) \otimes |\phi\rangle &= |\psi\rangle \otimes |\phi\rangle + |\psi'\rangle \otimes |\phi\rangle, \\
|\psi\rangle \otimes (|\phi\rangle + |\phi'\rangle) &= |\psi\rangle \otimes |\phi\rangle + |\psi\rangle \otimes |\phi'\rangle, \\
(\alpha|\psi\rangle) \otimes |\phi\rangle &= |\psi\rangle \otimes (\alpha|\phi\rangle) = \alpha\big(|\psi\rangle \otimes |\phi\rangle\big),
\end{aligned}$$

and similarly for duals.

The inner product on V satisfies

$$\big(\langle\psi'| \otimes \langle\phi'|\big)\big(|\psi\rangle \otimes |\phi\rangle\big) = \langle\psi'|\psi\rangle\langle\phi'|\phi\rangle,$$

extended sesquilinearly to arbitrary states (conjugate linear in the first state and linear in the second).

Given operators A on V_1 and B on V_2, define an operator $A \otimes B$ on V by

$$(A \otimes B)\big(|\psi\rangle \otimes |\phi\rangle\big) = (A|\psi\rangle) \otimes (B|\phi\rangle),$$

and extend by linearity. In particular,

$$A \longleftrightarrow A \otimes 1 \quad \text{acting only on } V_1, \qquad B \longleftrightarrow 1 \otimes B \quad \text{acting only on } V_2.$$

Operators acting on different factors commute for any A, B:

$$[A \otimes 1, 1 \otimes B] = 0.$$

Example: a particle in two dimensions

Consider the position operators $\hat{x}_1$, $\hat{x}_2$. A basis of joint eigenstates is constructed as

$$\underbrace{|x_1, x_2\rangle}_{\in V} = \underbrace{|x_1\rangle}_{\in V_1} \otimes \underbrace{|x_2\rangle}_{\in V_2},$$

where V_1, V_2 are the state spaces for a particle in one dimension. The position operators are identified with

$$\hat{x}_1 \longleftrightarrow \hat{x}_1 \otimes 1, \qquad \hat{x}_2 \longleftrightarrow 1 \otimes \hat{x}_2.$$

The wavefunction for a product state $|\psi\rangle \otimes |\phi\rangle$ is given by

$$\begin{aligned}
\langle x_1, x_2|\big(|\psi\rangle \otimes |\phi\rangle\big) &= (\langle x_1| \otimes \langle x_2|)(|\psi\rangle \otimes |\phi\rangle) \\
&= \langle x_1|\psi\rangle\langle x_2|\phi\rangle \\
&= \psi(x_1)\phi(x_2).
\end{aligned}$$

Thus

tensor product states $\longleftrightarrow$ separable wavefunctions,

tensor product space $\longleftrightarrow$ all linear combinations of such wavefunctions.

Note. If $\{|n\rangle\}$ is a basis for V_1 and $\{|m\rangle\}$ is a basis for V_2, then

$$\{|n\rangle \otimes |m\rangle\}$$

is a basis for V. If both bases are finite,

$$\dim V = (\dim V_1)(\dim V_2).$$

End of lecture 8